

MEASUREMENT & DIGITAL CONTROL



Week 3: Signal Conditioning

Agenda

0

Review & Motivation

Connecting Lectures 1 & 2 to today

2

Signal Generation

Resistive sensors - Wheatstone bridge - Quarter, half & full bridge

3

Signal Amplification

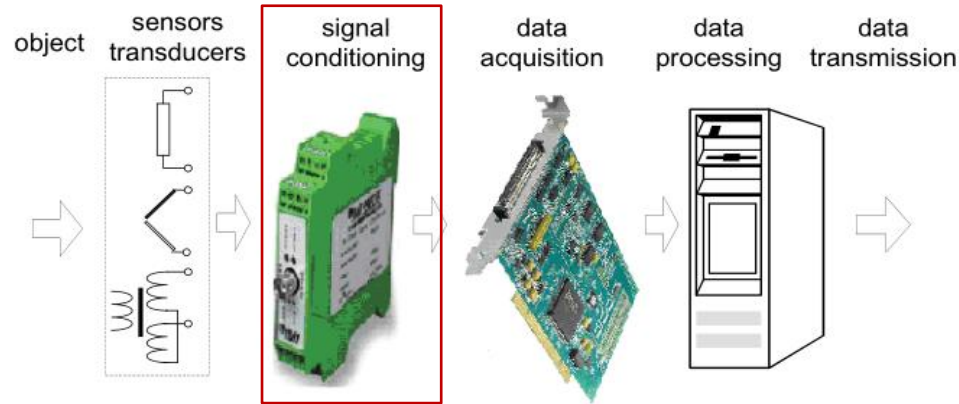
Loading - Buffers - In-Amp - CMRR - GBP - Phase distortion

4

Signal Filtering

Filter types - Filter families - Passive & active filter design

Review — Where are we?



Lecture 1

Errors, calibration & uncertainty

Architecture of measurement systems

Lecture 2

Dynamic characteristics

Transient response, bandwidth & phase lag, sensor selection

Lecture 3

Signal conditioning

The gap between sensor & DAQ

Signal conditioning — core functions

Excitation	Impedance conversion	Amplification	Filtering	Isolation
Supplies reference voltage or current to passive sensors (bridge, RTD, thermistor)	R→V (bridge circuit) I→V (transimpedance amp) V→I (4–20 mA current loop)	Scales mV sensor signal up to the 0–5 V range expected by the ADC	Removes noise & prevents aliasing before the ADC. Anti-aliasing filter = mandatory	Galvanic separation for safety & common-mode noise rejection

01

Signal Generation

Resistive sensors - Wheatstone bridge - Quarter, half & full bridge

The problem with resistive sensors

Examples: strain gauges, RTDs, thermistors, etc.

They change resistance, not voltage

- > ADCs read voltage — they cannot read resistance directly
- > ΔR is microscopic: for $GF = 2$, strain $\epsilon = 1000 \mu\epsilon$, $R = 120 \Omega$: $\Delta R = GF \times \epsilon \times R = 0.24 \Omega$
- > Measuring a 0.24Ω change in 120Ω by subtraction is impractical (0.2% precision) with standard ohmmeters
- > We need a circuit that outputs a voltage directly proportional to ΔR

Direct measurement (flawed)

Measure R before load: 120.00Ω
Measure R after load: 120.24Ω
Subtract $\rightarrow 0.24 \Omega$

Instrument resolution limits this to $\sim 0.1 \Omega \rightarrow$ huge relative error

Bridge measurement (correct)

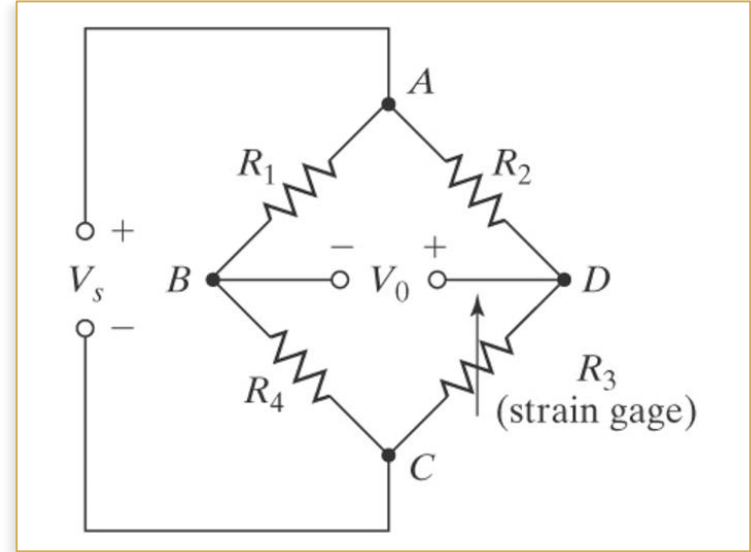
Set up a balanced bridge: $V_{out} = 0$
Apply load $\rightarrow$ bridge unbalances
Measure V_{out} directly

Small $\Delta R \rightarrow$ small V_{out} , easily amplified $\rightarrow$ high sensitivity & resolution

Quarter-bridge derivation (step-by-step)

Directly apply Kirchoff's law

- > $V_o = V_D - V_B$ (output is the potential difference between bridge midpoints)
- > $V_D = V_s \times R_2 / (R_2 + R_3)$ [voltage divider, right arm]
- > $V_B = V_s \times R_4 / (R_1 + R_4)$ [voltage divider, left arm]
- > Balanced state: $V_o = 0$ when $R_2/R_3 = R_1/R_4$



Key insight: When the bridge is balance, any change in R_3 unbalances the bridge, producing a measurable V_{out} proportional to ΔR .

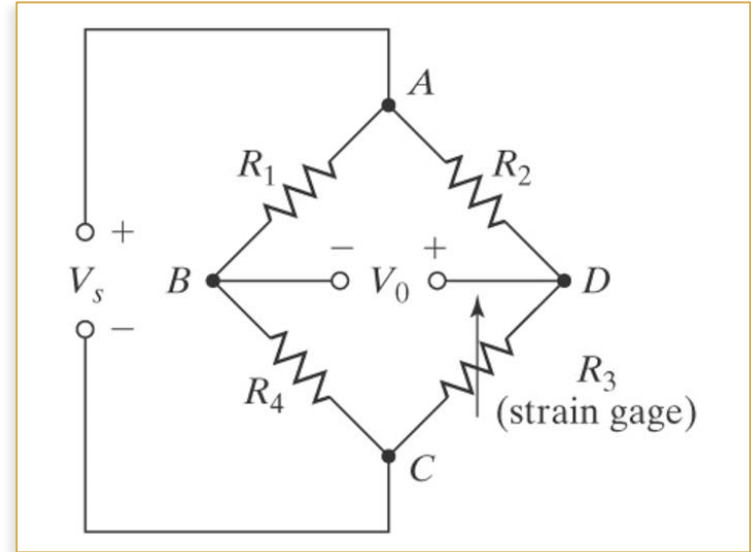
Quarter-bridge under load

Let the strain gage is strained: $R_3 = R_{3i} + \Delta R_3$

> Substituting into $V_o = V_D - V_B$ and simplifying

$$V_o = \left(\frac{R_3}{R_3 + R_2} - \frac{R_4}{R_1 + R_4} \right) V_s = \frac{V_s R_1 \Delta R_3}{(R_2 + R_{3i} + \Delta R_3)(R_1 + R_4)}$$

- > The ΔR in the denominator means output is NOT a perfectly straight line vs. strain [non-linear]
- > In practice for metal gauges: $\Delta R/R \approx 0.002$ at $1000 \mu\epsilon \rightarrow$ denominator $\approx 1.001 \rightarrow$ error $< 0.1\%$
- > But the nonlinearity is real and matters for large strains or precision metrology



Fixing the flaw — half-bridge and full-bridge

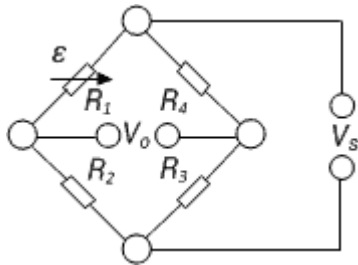
Quarter-bridge

1 active gauge: $V_o = \frac{(V_s/4)/(\Delta R/R)}{1+(\Delta R/2R)}$

Nonlinear ΔR term in denominator

Sensitivity: 1×

No temperature compensation

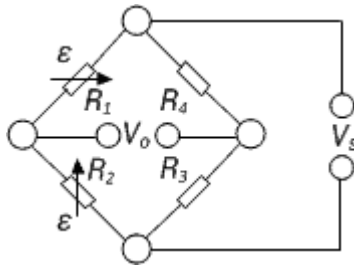


Half-bridge

2 active gauges (+ ΔR tension, $-\Delta R$ compression)

Purely linear: $V_o = \frac{V_s \Delta R}{2R}$

Sensitivity: 2×



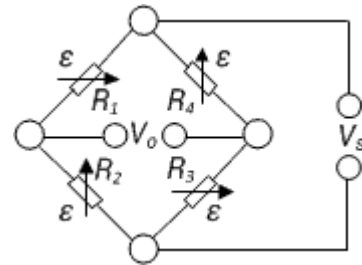
Full-bridge

2 active gauges (2 tension, 2 compression)

Purely linear: $V_s \Delta R/R$

Sensitivity: 4×

Temperature self-compensation



Example

Given: Full-bridge strain gauge, $V_s = 10\text{ V}$, Gauge Factor $GF = 2.0$, Applied strain $\epsilon = 1000\text{ }\mu\epsilon$. How small is the signal we need to amplify, what's the amplifier's gain?

1. $\Delta R/R = GF \times \epsilon = 2.0 \times 0.001 = 0.002$
2. Full-bridge output: $V_o = V_s \times (\Delta R/R) = 10 \times 0.002 = 0.020\text{ V} = 20\text{ mV}$
3. ADC requirement: $0 - 5\text{ V} \rightarrow \text{Required gain} = 5\text{ V} / 0.020\text{ V} = 250\times$

20 mV

That is our signal. The ADC expects 0–5 V. We need a gain of $250\times$ — safely, without distorting the tiny differential voltage the bridge produces.

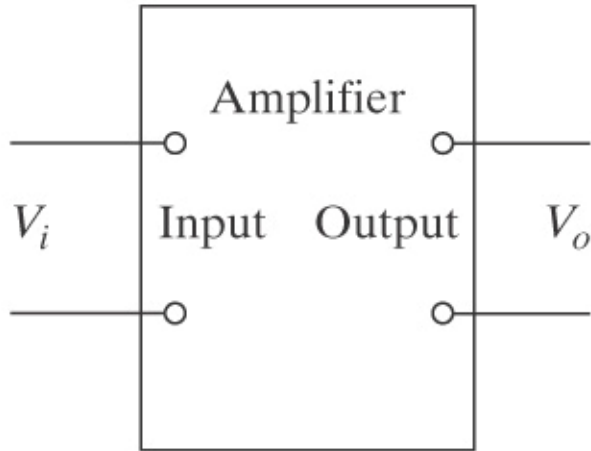
How do we get there?

02

Amplification & Loading Effects

Loading · Buffers · In-Amp · CMRR · GBP · Phase distortion

The amplification problem



Obstacle 1 — loading effects

Bridge-amplifier interface: forms a voltage divider (amplifier's Z_{in} vs. bridge's Z_{out})

If $Z_{in} \sim Z_{out}$, signal attenuates before amplification → loading error

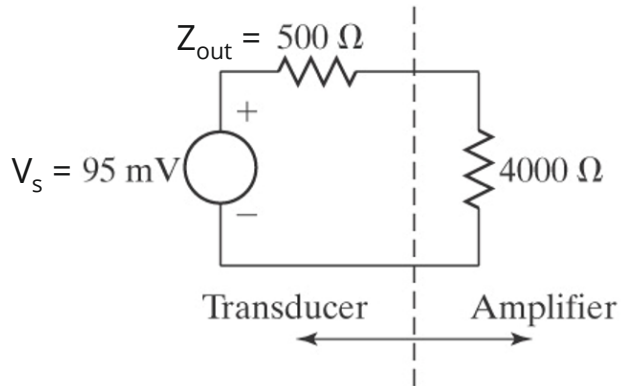
Obstacle 2 — common-mode noise

In real environments, motors and cables induce 50/60 Hz noise equally on both bridge outputs (common-mode noise)

Basic inverting amplifiers would amplify both signal and noise → use a *differential amplifier* to amplify only the difference

Both obstacles are solved by the same device: the Instrumentation Amplifier (In-Amp). But we need to understand loading and differential measurement first to appreciate why.

Loading effects



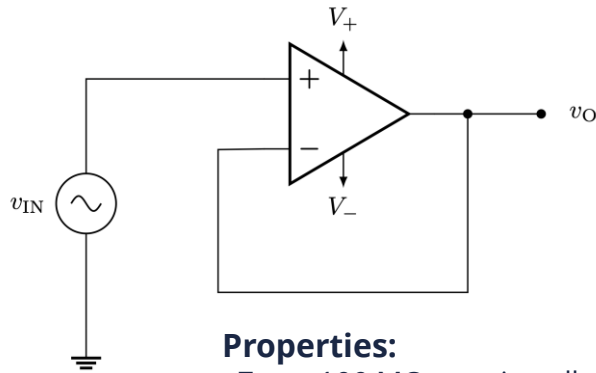
- Sensor has output impedance Z_{out} , amplifier has input impedance Z_{in}
- The voltage actually reaching the amplifier input:

$$V_{\text{measured}} = V_s \times Z_{in} / (Z_{in} + Z_{out})$$

Z_{in}	Vmeasured ($V_s = 95 \text{ mV}$, $Z_{out} = 500 \Omega$)	Error
4 k Ω	$95 \times 4000 / 4500 = 84.4 \text{ mV}$	11.2%
50 k Ω	$95 \times 50000 / 50500 = 94.1 \text{ mV}$	1.0%
1 M Ω	$95 \times 1\text{M} / 1.0005\text{M} = 94.95 \text{ mV}$	0.05%

Engineering rule: $Z_{in} \geq 100 \times Z_{out} \rightarrow \text{loading error} < 1\%$ (Amplifier input impedance must dominate $\rightarrow$ this is why we use op-amp input buffers - they present $Z_{in} \approx 100 \text{ M}\Omega$ to the sensor)

Solving loading — the voltage follower (buffer)



Properties:

$Z_{in} \approx 100 \text{ M}\Omega \rightarrow$ virtually no loading

$Z_{out} \approx 1 \text{ }\Omega \rightarrow$ drives any load

$G = 1 \rightarrow$ signal unchanged

Why this matters for bridges

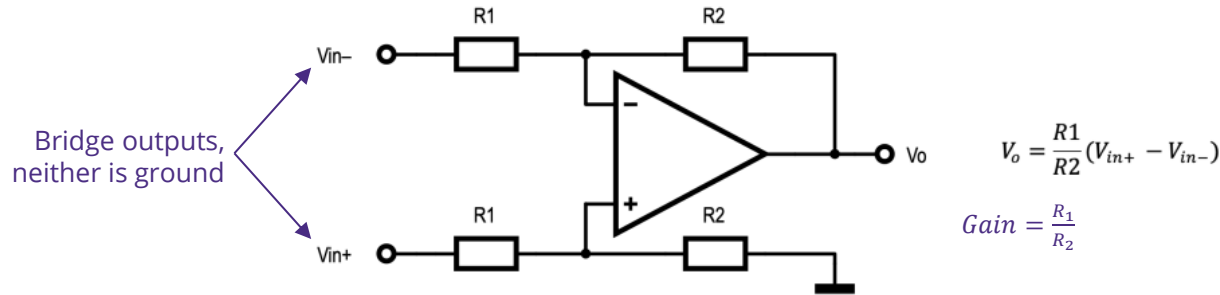
A Wheatstone bridge has $Z_{out} \approx R/2$ (e.g. $60 \text{ }\Omega$ for $120 \text{ }\Omega$ gauges).

Connecting a $1 \text{ M}\Omega$ buffer eliminates loading error to $< 0.006\%$.

Key insight: op-amps are used not just to multiply signals - also to protect and isolate them.

However: a bridge outputs a difference between two floating nodes - neither V_A nor V_B is ground. A single buffer doesn't solve this $\rightarrow$ need differential measurement.

The differential measurement problem



Standard diff-amp: the fatal flaw

Input impedance is determined by the resistor network.
For $R1 = 10\text{ k}\Omega$ resistors: $Z_{in} \approx 20\text{ k}\Omega$

Connecting this to a $120\text{ }\Omega$ bridge: loading error =
 $120/(120+20000) \approx 0.6\%$

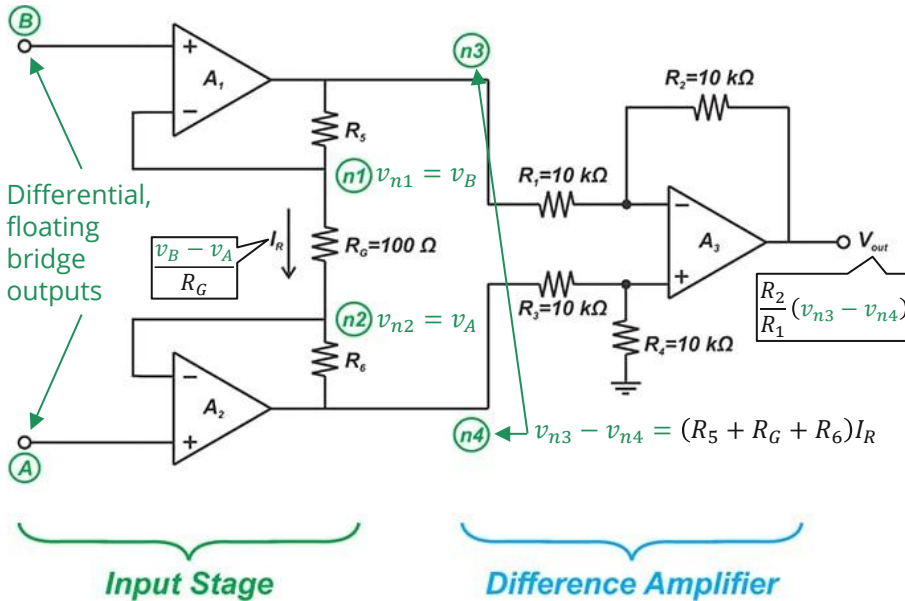
Worse: unequal loading on V_A and V_B causes gain error
AND CMRR degradation.

What we actually need

1. Infinite Z_{in} on both V_{in-} & $V_{in+} \rightarrow$ zero loading
2. Measure only V_{in-} and V_{in+} , reject any signal common to both
3. High, stable gain

Solution: the Instrumentation Amplifier

Instrumentation Amplifier (In-Amp)



Why Instrumentation Amplifier?

Very high $Z_{in} \rightarrow$ no loading of the bridge

Differential only $\rightarrow$ rejects common-mode noise

Provides accurate, stable gain set by a single resistor: $G = 1 + \frac{K}{R_G}$

Real devices: AD620

Single 8-pin IC, gain set with one resistor.

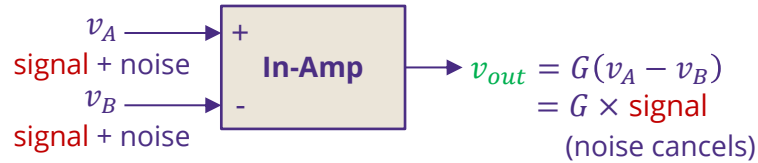
$K = 49.4\ k\Omega \Rightarrow$ for $G = 250$, R_G must be set to $200\ \Omega$

Input $Z_{in} > 10\ G\Omega$

CMRR = 100 dB

💡 Instrumentation amplifiers enable accurate, low-noise measurement of small differential bridge signals

Common-mode Signal Rejection with In-Amp



Definition

$CMRR = 20 \log_{10}(A_d/A_{cm})$ [dB], where A_d = differential gain, A_{cm} = common-mode gain.

Ability to reject common-mode signals.

AD620 example

signal = 1 mV, noise = 1 V, Gain = 100, CMRR = 100 dB.

$CMRR = 20 \log_{10}(100/A_{cm}) = 100 \text{ dB} \Rightarrow A_{cm} = 0.001 \Rightarrow$
Output noise $\approx 0.001 \times 1 \text{ V} = 1 \text{ mV}$.

Output signal = $100 \times 1 \text{ mV} = 0.1 \text{ V} \Rightarrow$ signal clearly dominates after amplification

Why it matters

Rejects 50/60 Hz noise

Critical for:

- Strain gauges
- Biomedical signals

💡 High CMRR allows extraction of tiny signals buried in large noise

Real op-amp: Gain-bandwidth product

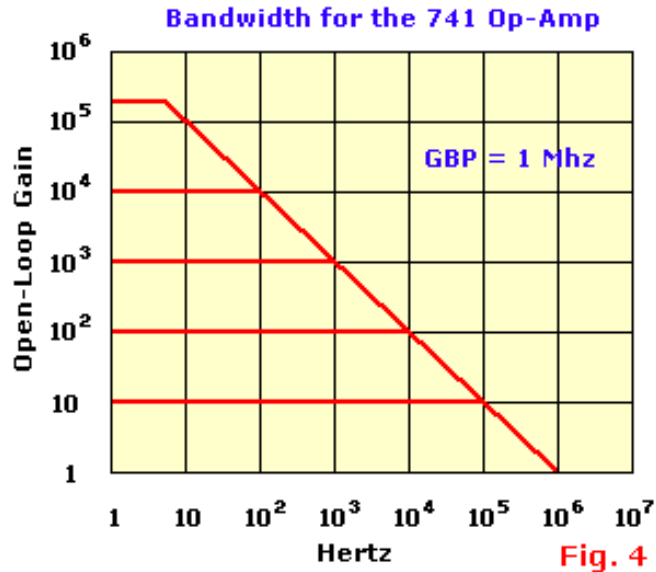


Fig. 4

- > Ideal op-amps have infinite bandwidth, *real* op-amps do not
- > Internally compensated op-amps: $\text{Gain} \times \text{Bandwidth} = \text{const}$
- > As gain *increases*, the usable BW *shrinks proportionally*

Example: A sensor outputs a 5 mV (peak) signal at 20 kHz. Can an op-amp with 1 MHz GBW amplify it to 2 V (peak)?

Answer: No, the op-amp *cannot* faithfully reproduce the signal

- Required gain: $2 \text{ V} / 5 \text{ mV} = 400$
- Available BW: $1 \text{ MHz} / 400 = 2.5 \text{ kHz}$

Engineering fix: Use either

- An op-amp with GBW of at least $400 \times 20 \text{ kHz} = 8 \text{ MHz}$; or
- A 2-stage amplifier (each stage has $G = 20$ & $BW = 50 \text{ kHz}$).

03

Signal Filtering

Why filter · Filter types · Filter families · Passive & active filter design

Why filter?

Remove existing noise

- > Even with high CMRR, high-frequency electrical noise remains after amplification.
- > Sources: switching power supplies, RF interference, cable pickup.
- > Solution: low-pass filter removes all noise above the signal bandwidth.

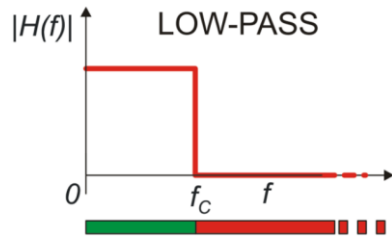
Prevent aliasing

- > **Aliasing:** frequencies above half of the sampling frequency ($f_s/2$) fold into lower frequencies
- > This damage is permanent - no digital processing can undo it.
- > Solution: anti-aliasing filter removes all content above $f_s/2$ before sampling.

💡 **No analog filter, no protection:** *aliasing must be prevented before sampling, not after.*

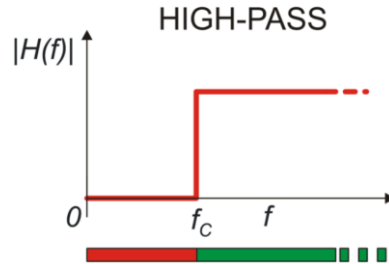
Ideal analog filters

Low-pass filter



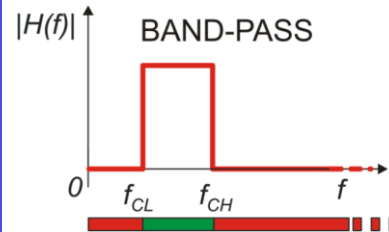
- > Passes DC to f_c , rejects high-frequency noise
- > Most common in measurement
- > Anti-aliasing filter type

High-pass filter



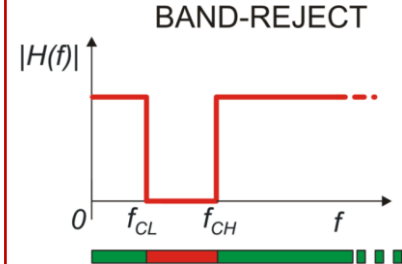
- > Rejects DC offset
- > Passes above f_c
- > Used to remove drift or baseline wander

Band-pass filter



- > Passes a frequency band between f_{CL} and f_{CH}
- > Used in vibration analysis

Band-reject filter



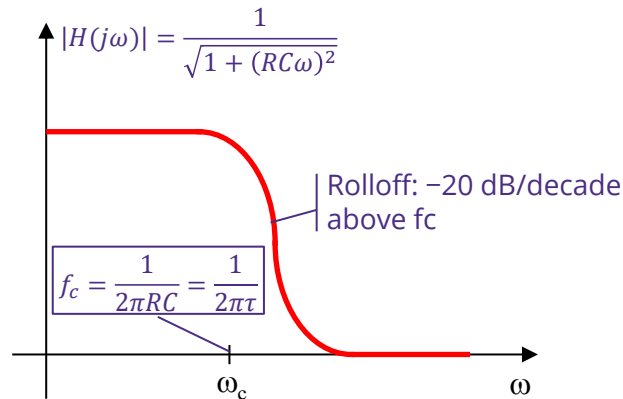
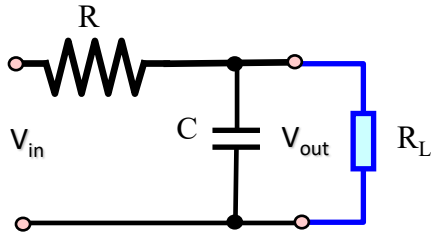
- > Rejects a specific band, e.g., 50 Hz notch filter
- > Targets a known interference frequency

Real analog filters

Family	Passband	Phase	Rolloff	Best for
Butterworth	Maximally flat	Good	Moderate	General measurement — flat gain more important than sharp cutoff
Bessel	Slight droop	Linear (best)	Gentle	Pulse & vibration — waveform shape must be preserved
Chebyshev	Equi-ripple	Fair	Steep	Separation of closely-spaced frequency bands
Elliptic	Equi-ripple	Poor	Steepest	Hardware-constrained — minimum component count

💡 *Default for most measurement: Butterworth (flat passband, good phase). Switch to Bessel when waveform shape must be preserved.*

Passive RC low-pass filter



Design procedure

1. Select resistor based on output loading limit, e.g. 5% $\Rightarrow R \approx 0.05 \times R_L$
2. Select capacitor based on cut-off frequency: $C = \frac{\tau}{R} = \frac{1}{2\pi R f_c}$

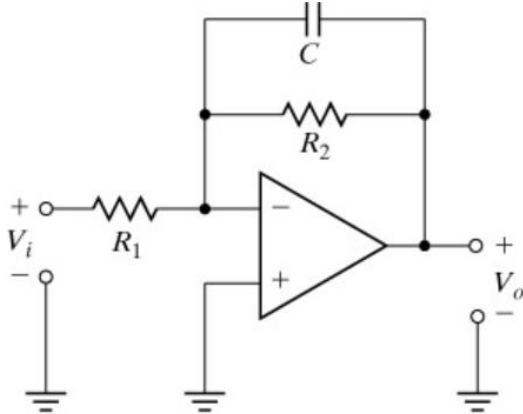
Example: Pressure sensor outputs signal to 3 Hz, 60 Hz noise, next stage's input resistance: 10 k Ω .

In order to achieve $\leq 5\%$ loading, $R \leq 0.05 \times R_L = 500 \Omega \rightarrow$ choose standard value: $R = 470 \Omega$.

$f_c = 3 \text{ Hz} \Rightarrow C = \frac{1}{2\pi R f_c} \approx 113 \mu\text{F} \rightarrow$ choose $C = 100 \mu\text{F}$

Attenuation at 60 Hz: $20 \log_{10} \frac{1}{\sqrt{1 + (f/f_c)^2}} \approx -24.6 \text{ dB}$, which might not be sufficient for anti-aliasing.

Active filter design



Design example: Pressure sensor outputs signal to 3 Hz, 60 Hz noise, next stage's input resistance: 10 k Ω

Filter choice:

1. No pulse-shape requirement: Bessel is not needed ✗
2. Butterworth: flat passband, good phase ✓

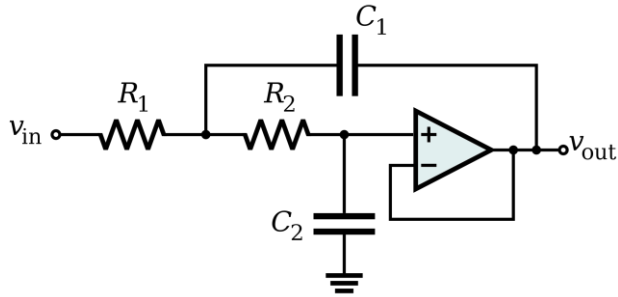
Circuit design:

1. No gain needed $\rightarrow$ unity gain $\rightarrow R_1 = R_2 = 10 \text{ k}\Omega$
2. $f_c = 3 \text{ Hz} \rightarrow C = 1/(2\pi \times 10\text{k} \times 3) = 5.3 \mu\text{F}$

Verification:

1. Octaves from f_c to noise: $2^n = 60/3 \rightarrow n = \log_2(20) = 4.3$
2. Attenuation at 60 Hz (-6 dB/octave): $-6 \times 4.3 = -25.8 \text{ dB}$
3. If -25.8 dB is insufficient: use 2nd-order filter (-40 dB/decade)

Higher-order filters & the Sallen-Key topology



Higher-order filters

When a 1st-order filter is not enough

- 2nd order: -40 dB/decade → standard anti-aliasing filter
- 4th order: -80 dB/decade → high-precision instruments

Sallen-Key 2nd-order active filter

- > Most common active filter topology in measurement — 1 op-amp, 2 resistors, 2 capacitors
- > Configurable for any family (Butterworth, Bessel, Chebyshev) by changing component ratios
- > Butterworth equal-component design (maximally flat): $R_1 = R_2 = R$, $C_1 = C$, $C_2 = C/\sqrt{2}$
- > Cutoff frequency: $f_c = 1 / (2\pi R \sqrt{C_1 \times C_2})$

KEY TAKEAWAYS

1 Bridge circuits

Full bridge: 4× sensitivity, linear, temperature-compensated.

2 Loading

$Z_{in} \geq 100 \times Z_{out} \rightarrow \text{error} < 1\%$. Use op-amp buffers for high-impedance sources ($Z_{in} \approx 100 \text{ M}\Omega$).

3 In-Amp

Correct tool for differential + noisy environments. $\text{CMRR} \geq 80 \text{ dB}$; gain set by one resistor R_G .

4 GBP

$\text{GBP} = G \times f_c$. Higher gain costs bandwidth. Plan gain budget against highest signal frequency.

5 Anti-aliasing

Mandatory analogue LPF before every ADC

6 Filter choice

Flat gain $\rightarrow$ Butterworth; Linear phase $\rightarrow$ Bessel; Sharp rolloff $\rightarrow$ Chebyshev/Elliptic

Next Lecture: Data Acquisition